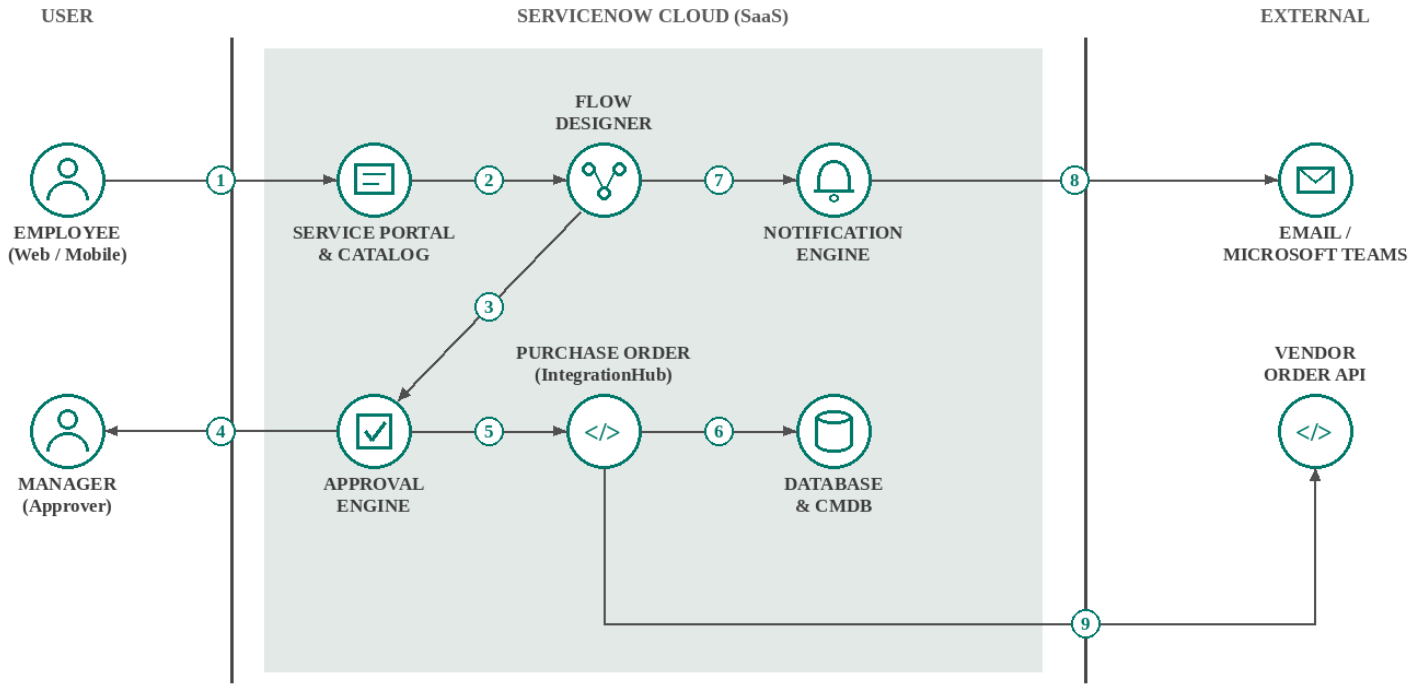
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 September 2026
Team ID	SWTID-2026-1286
Project Name	Streamlining IT Procurement
Maximum Marks	4 Marks

Streamlining IT Procurement: Automating Standard Laptop Orders with IT Procurement

The deliverable includes the architectural diagram below and the information as per Table-1 and Table-2. The architecture shows how a standard laptop request raised by an employee flows through the service portal, approval workflow, vendor order integration and asset inventory.

Architecture: Automating standard laptop orders in IT procurement



1. Employee submits a standard laptop request through the service portal.
2. The request triggers a flow in Flow Designer.

3. The flow sends the request to the approval engine.
4. The manager approves or rejects the request.
5. An approved request moves to purchase order creation.
6. Order details are stored in the database and CMDB.
7. The flow triggers the notification engine.
8. Email / Teams notifications are sent to the employee and manager.
9. The purchase order is sent to the vendor order API.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Employees, managers and procurement staff interact with the application through a web portal and mobile app	ServiceNow Service Portal (HTML, CSS, JavaScript, AngularJS), ServiceNow Mobile App
2.	Application Logic-1	Standard laptop request through a service catalog item with predefined specifications	ServiceNow Service Catalog, Record Producer (JavaScript)
3.	Application Logic-2	Automated approval workflow that routes each request to the reporting manager, with reminders	Flow Designer, ServiceNow Approval Engine
4.	Application Logic-3	Automatic purchase order creation and vendor order placement for approved requests	Flow Designer, IntegrationHub (REST)
5.	Database	Stores requests, approvals, purchase orders and laptop catalogue data	ServiceNow platform database (relational)
6.	Cloud Database	Asset and configuration records updated when a laptop is delivered	ServiceNow CMDB and Asset Management
7.	File Storage	Attachments such as quotations, invoices and delivery notes	ServiceNow attachment storage (cloud)
8.	External API-1	Vendor API to place laptop orders and receive shipment status	Vendor REST API through IntegrationHub
9.	External API-2	Email and chat notifications to employees and managers on every status change	SMTP email, Microsoft Teams API
10.	Machine Learning Model	Not required for the core solution. Optionally used to forecast laptop demand	ServiceNow Predictive Intelligence (optional)
11.	Infrastructure (Server / Cloud)	Application deployment on Cloud Local Server Configuration: Not required (browser-based access) Cloud Server Configuration: ServiceNow cloud instance	ServiceNow Cloud (SaaS)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks used to build portal widgets and the user interface	AngularJS, Bootstrap
2.	Security Implementations	Single sign-on, role-based access control, encrypted communication and audit logs for approvals and orders	SAML SSO, ACLs and roles, HTTPS/TLS, audit logs, OWASP guidelines
3.	Scalable Architecture	Multi-tier cloud architecture (presentation, application and database) that scales with the number of employees and orders	3-tier architecture on ServiceNow cloud
4.	Availability	Hosted on cloud with redundant data centres, load-balanced application nodes and automatic failover	ServiceNow multi-instance architecture, load balancers, disaster recovery
5.	Performance	Automated flows process requests within seconds; vendor calls run asynchronously and frequently used data is cached	Flow Designer asynchronous execution, platform caching, CDN